

UAVs / Aeronaves Robotizadas - Course Knowledge Base (2022/23)

Course notes – Autumn 2022/23

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This file is the entry point into a deep technical analysis of every lecture PDF in this folder. Detailed, equation-by-equation notes for each group of chapters live in notes/ (linked below). This file gives the course overview, the chapter map, and how the pieces fit together.

1 Course identity

- **Course:** Unmanned Aerial Vehicles / Aeronaves Robotizadas (ARob), MEAer, Instituto Superior Técnico, Lisboa.
- **Edition:** Autumn Semester 2022/2023.
- **Faculty:** José Raul Azinheira (jraz@dem.ist.utl.pt, responsible), Rita Cunha (rita@isr.tecnico.ulisboa.pt).
- **Structure:** 2h/week lectures + 1h30/week lab (12h total), 3 lab assignments (L1 modelling/identification, L2 motion-variable estimation, L3 motion control), 1 report per lab.
- **Grading:** $E = \max(E1, E2)$ (exam, 40%), $L = 0.2L1 + 0.4L2 + 0.4L3$ (labs, 60%), $F = 0.4E + 0.6L$.
- **Main textbook:** Beard & McLain, *Small Unmanned Aircraft: Theory and Practice*, Princeton University Press, 2012 - directly reused (sometimes verbatim) for Ch5, Ch6, Ch8, Ch9.
- **Other references:** Astrom & Murray, *Feedback Systems*; Mahony/Kumar/Corke, "Multirotor Aerial Vehicles" (IEEE RAM 2012); Khalil, *Nonlinear Systems*; Slotine & Li, *Applied Nonlinear Control*; Lee/Leok/McClamroch, "Geometric tracking control of a quadrotor UAV on SE(3)" (CDC 2010); Cabecinhas/Cunha/Silvestre (2014, by the course's own instructor Rita Cunha).

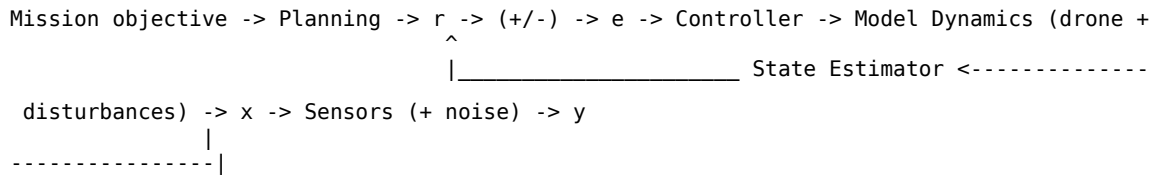
Important numbering gotcha: the course’s own chapter numbers (0-9) do **not** match Beard & McLain’s book chapter numbers. Course Ch5→book Ch7, course Ch6→book Ch8, course Ch8→book Ch10, course Ch9→book Ch11. ar23-ch10.pdf is *not* a 10th technical chapter - it’s an 11-page course wrap-up/conclusion talk that recaps the syllabus and lab exercises.

2 Chapter map and detailed notes

#	Topic	Source PDFs	Detailed notes
Ch0	Course intro, UAV motivation, vehicle-type tradeoffs, generic control-loop architecture	Ch0_Intro_Part_I/II	notes/01-foundations.md
Ch1	Rigid-body kinematics & dynamics (rotations, Newton-Euler)	Ch1_Rigid_Body	notes/01-foundations.md
Ch2	Quadrotor dynamic modelling, actuation, differential flatness, SWaP design	Ch2_Quadrotor_Modeling	notes/01-foundations.md
Ch3	Quadrotor control intro: cascade position/attitude control, root-locus/Bode loop-shaping, integral action	Ch3_Quadrotor_Control_Intro	notes/02-control.md
Ch4	Modern control design: controllability/observability, pole placement, Luenberger observers, LQR	Ch4_Modern_Control_Design	notes/02-control.md
Ch5	Sensors (accelerometer, gyro, magnetometer, barometer, GPS) + AR.Drone specifics (sonar, vision)	Ch5-6_Sensors_and_State_Estimation_1, Ch5_Sensors_Beard_McLain	notes/03-sensors-estimation.md
Ch6	State estimation: Kalman filter, EKF, complementary filters, GPS smoothing, measurement gating	Ch5-6_Sensors_and_State_Estimation_1, Ch6_State_Estimation_Beard_McLain	notes/03-sensors-estimation.md
Ch7	Nonlinear control: Lyapunov theory, LaSalle, backstepping, adaptive control, geometric SO(3) attitude control	Ch7_Intro_Nonlinear_Control, ..._Part_II	notes/04-guidance-nonlinear.md
Ch8	Path following: straight-line and orbit guidance laws (= book Ch.10)	Ch8_Path_Following	notes/04-guidance-nonlinear.md
Ch9	Path manager: waypoint switching, fillets, Dubins paths (= book Ch.11)	Ch9_PathManager	notes/04-guidance-nonlinear.md
(wrap-up)	Course conclusion talk, confirms syllabus/chapter numbering	ar23-ch10	notes/04-guidance-nonlinear.md

3 The big picture: how it all fits together

The course is organized around one recurring control-loop diagram (introduced in Ch0):



Every later chapter fills in one block of this diagram:

1. **Model Dynamics** (Ch1-Ch2): derive the general rigid-body equations of motion, then specialize them to the quadrotor. Result is the base plant used everywhere downstream: $\dot{p} = Rv$, $\dot{R} = RS(\omega)$, $m\dot{v} = -S(\omega)mv + f$, $J\dot{\omega} = -S(\omega)J\omega + n$, with quadrotor-specific $f = -Te_3 + mgR^Te_3$ and mixer matrix mapping 4 rotor thrusts to (T, n) . The quadrotor is **underactuated** (12 states, 4 inputs) but **differentially flat** in (p, ψ) - this flatness property is exactly what makes the cascade control design in Ch3/Ch7 possible (a desired thrust direction can always be recovered algebraically from a desired acceleration).
2. **Controller** (Ch3, Ch4, Ch7): three increasingly rigorous ways to design the same cascade (outer position loop $\rightarrow$ inner attitude loop) controller:
 - Ch3: classical SISO PD/PID + root-locus/Bode loop-shaping, treating the two loops as independent because the inner loop is designed 5-10x faster.
 - Ch4: the state-space/"modern" reformulation of the same problem - pole placement and LQR replace hand-tuned PID, and the same architecture (state feedback + observer) is shown to collapse the hierarchical structure into one full-state feedback law when desired.
 - Ch7: the nonlinear/Lyapunov-rigorous version of the *same* cascade - backstepping reproduces the Ch3 PD law but with a stability proof that doesn't require linearization, and geometric SO(3) control handles the attitude loop in a way that respects the topology of rotations (revealing the fundamental result that *global* attitude stabilization is impossible with continuous feedback).
3. **Sensors + State Estimator** (Ch5-Ch6): the accelerometer/gyro/magnetometer/GPS models feed a Kalman filter / EKF, whose gain-design math ($L = \Sigma C^T N^{-1}$, Riccati equation) is the exact **dual** of the LQR gain design from Ch4 ($K = R^{-1} B^T P$) - the course makes this LQR $\leftrightarrow$ KF duality explicit. The complementary filter used in the AR.Drone labs is shown to be a fixed-gain special case of the Kalman filter.
4. **Planning / Guidance** (Ch8-Ch9): sits *above* the control loop, generating the reference $r(t)$ that the controller tracks. Path following (Ch8) gives robust-to-wind vector-field laws for single line/orbit segments; the path manager (Ch9) sequences waypoints into continuous paths using half-plane switching, fillets, or Dubins paths, feeding its output as the position reference into the Ch3/Ch4/Ch7 controllers.

Course $\leftrightarrow$ lab mapping (from Ch0 logistics): L1 = Modelling/identification ($\rightarrow$ Ch1/Ch2), L2 = Estimation ($\rightarrow$ Ch5/Ch6), L3 = Motion control ($\rightarrow$ Ch3/Ch4/Ch7). The AR.Drone platform used in labs has **no GPS** - it substitutes sonar (height) and downward-camera vision (velocity via optical flow / corner tracking), which is why the generic Beard & McLain GPS-smoothing material (Ch6) doesn't directly

transfer to the labs and is explicitly flagged as "generic background" in the sensors notes.

4 Consolidated gotchas / things to double-check before relying on this material

- The quadrotor mixer matrix M sign/index convention (which rotor pair drives n_x vs n_y) is convention-dependent and left as an open in-class exercise for the $\times$ configuration and hexarotor case - don't trust a specific sign pattern without checking the rotor-numbering diagram in use.
- Ch3's inner-loop pole-placement example references gains k_ω , k_λ symbolically without stating numeric values on the slides.
- There's a typo in the Ch5-6 course slides: the stochastic output equation is written $y = Cy + n$ but should read $y = Cx + n$.
- The LQR $\leftrightarrow$ Kalman-filter duality slide leaves "(A, $\tilde{B}$) controllable?" as an open question mark in the source material itself.
- The geometric $SO(3)$ attitude-control derivation (Ch7 Part II) ends mid-derivation on the exact constant needed for local (as opposed to almost-global) stability - referred to Lee/Leok/McClamroch (2010) rather than spelled out.
- Quaternions are mentioned once (Ch1) as an alternative attitude representation but never developed - if a lab or exam question needs quaternion kinematics, it's not in these slides.

See each notes/0N-*.md file for full equation-by-equation detail, worked examples, and chapter-specific open questions.